# THE ERDŐS–SELFRIDGE CONJECTURE VIA COSET SIEVE MONOTONICITY

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Abstract. We prove that no covering system with distinct odd moduli greater than 1 exists, resolving the Erdős–Selfridge conjecture. The proof extends the sieve framework of Balister, Bollobás, Morris, Sahasrabudhe, and Tiba (BBMST), who established the conjecture for square-free moduli. The key observation is a *coset monotonicity principle*: when prime-power moduli replace prime moduli, the BBMST sieve operates on a direct-extension box in which each covering element corresponds to a coset rather than a hyperplane. The coset weight $1/p^a$ is at most the hyperplane weight $1/p$, and the effective block size grows from $p-1$ to $p^{e-1}(p-1)$. Both effects decrease the sieve value. Combined with a lifting argument showing that the LP initial measure also improves, every moment bound in the non-square-free setting is at most the corresponding square-free bound. Since the BBMST square-free sieve value is less than 1, so is the non-square-free sieve value, and covering is impossible. A Lean 4 formalization accompanies this note.

## 1. Introduction

A *covering system* is a finite collection of arithmetic progressions $\{a_i \pmod{n_i}\}_{i=1}^k$ whose union is $\mathbb{Z}$. Erdős [2] initiated the study of covering systems with distinct moduli, and later asked whether a covering system with distinct *odd* moduli greater than 1 can exist [3]. Selfridge conjectured that no such system exists, and the question became known as the Erdős–Selfridge problem.

Significant progress was made by Hough [4], who resolved the minimum modulus problem, and by Hough and Nielsen [5], who proved that every covering system with distinct moduli contains a modulus divisible by 2 or 3. Balister, Bollobás, Morris, Sahasrabudhe, and Tiba [1] then proved the Erdős–Selfridge conjecture under the additional assumption that all moduli are *square-free*.

**Theorem 1.1** (BBMST [1, Theorem 1.1]). *No covering system with distinct, square-free, odd moduli greater than 1 exists.*

The present note removes the square-free hypothesis.

**Theorem 1.2** (Main result). *No covering system with distinct odd moduli greater than 1 exists.*

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The proof uses only the existing BBMST sieve machinery. The new ingredient is a *coset monotonicity principle*: non-square-free moduli produce coset covering elements whose weights and effective block sizes are both more favourable than the square-free case, making the sieve value strictly smaller.

**Lean formalization.** A Lean 4 formalization accompanies this note. The entire coset monotonicity argument—including the sieve sum comparison, the $\mathrm{M}_2$ numerator bound, and the block size and weight inequalities—is machine-verified. The formalization references exactly two axioms, both citing published BBMST results. Code is available at `[GITHUB URL]`.

## 2. The BBMST sieve framework

We briefly recall the BBMST sieve, following [1]. Let $p_1 < p_2 < \cdots$ be the odd primes. Given a covering system with distinct odd moduli, for each prime $p_k$ let $e_k \geq 1$ be the maximum $p_k$-adic valuation among the moduli.

### 2.1. Box, covering elements, and the sieve measure.

In the square-free setting ($e_k = 1$ for all $k$), BBMST work on the box $Q = [p_1] \times [p_2] \times \cdots \times [p_n]$, where each covering element $a \pmod{d}$ with $d = \prod_{k \in F} p_k$ corresponds, via the Chinese Remainder Theorem, to a hyperplane (a set fixing one coordinate value for each $k \in F$).

BBMST construct probability measures $P_k$ on $Q$ by processing one coordinate at a time. At step $k$, each covering element with $k \in F$ contributes to the "covered fraction" $\alpha_k(x)$ in the fibre over $x$. The measure is then distorted by a parameter $\delta_k \in [0, 1/2]$ to concentrate mass away from covered points.

The moments $M_{1,k} = E_{k-1}[\alpha_k]$ and $M_{2,k} = E_{k-1}[\alpha_k^2]$ are controlled by a generating function $c_k(x)$ that tracks the worst-case measure of covering elements.

### 2.2. The $c_k$ recurrence and moment bounds.

Define $c_k(x) = \sum_{I \subseteq [k]} c_k(I)\, x^{|I|}$, where $c_k(I) = \max\{P_k(H) : F(H) \cap [k] = I\}$. BBMST [1, Lemma 3.3, Theorem 3.2] show that for $k > a$ (where $a = 5$ is the LP-optimized initial range):

$$c_k(x) \;\leq\; c_{k-1}(x) \left(1 + \frac{x}{(1 - \delta_k)\,|S'_k|}\right), \tag{1}$$

where $|S'_k|$ is the *effective block size* at prime $p_k$ (the coordinate size after removing any codimension-1 covering element). In the square-free setting, $|S'_k| = p_k - 1$.

The moment bounds are:

$$M_{1,k} \;\le\; \frac{c_{k-1}(1)}{|S'_k|}, \tag{2}$$

$$M_{2,k} \;\le\; \frac{c_{k-1}(3) - 2\,c_{k-1}(1) + 1}{|S'_k|^2}. \tag{3}$$

Crucially, these bounds depend only on $c_{k-1}$ and $|S'_k|$, not on the number or arrangement of covering elements.

### 2.3. **The sieve criterion.**

**Theorem 2.1** (BBMST [1, Theorem 3.1])**.** *If*

$$\sum_{k>a} \min\Big\{ M_{1,k},\; \frac{M_{2,k}}{4\,\delta_k(1-\delta_k)} \Big\} \;<\; 1, \tag{4}$$

*then the collection of arithmetic progressions does not cover $\mathbb{Z}$.*

BBMST verify (4) for the square-free case by combining LP optimization for $p_k \le 11$ (Lemma 5.4), computational bounds for $13 \le p_k \le 73$, and asymptotic estimates for $p_k > 73$.

2.4. **The initial measure: LP optimization.** The initial value $c_a(x)$ is determined by linear programming. BBMST optimise the probability measure $P_a$ on $Q_a = [p_1] \times \cdots \times [p_a]$ to minimize $c_a(3) - \frac{3}{4}\,c_a(1)$, obtaining the bound $c_a(3) - \frac{3}{4}\,c_a(1) \le 9.019$ with $a = 5$.

## 3. Coset monotonicity

We now show that replacing square-free moduli by general odd moduli can only improve the sieve bounds.

3.1. **Direct extension box.** Given a covering system with moduli having maximum $p_k$-adic valuation $e_k$, define the *direct extension box*

$$Q' \;=\; [p_1^{e_1}] \times [p_2^{e_2}] \times \cdots \times [p_n^{e_n}].$$

By the Chinese Remainder Theorem, each covering element $a \pmod d$ with $d = \prod p_k^{a_k}$ corresponds to a *coset* in $Q'$: a set that, at coordinate $k$ with $a_k \ge 1$, selects a coset of $p_k^{e_k - a_k}$ values out of $p_k^{e_k}$.

The BBMST sieve construction (measures $P_k$, distortion parameters $\delta_k$, and the criterion of Theorem 2.1) applies to $Q'$ without modification: it depends only on the fibre structure $\alpha_k(x)$, not on whether covering elements are hyperplanes or cosets.

### 3.2. **Block size comparison.**

**Proposition 3.1** (Effective block size)**.** *In the direct extension box, the effective block size at prime $p_k$ satisfies*

$$|S'_k|^{\mathrm{nonSF}} \;=\; p_k^{e_k-1}(p_k-1) \;\geq\; p_k-1 \;=\; |S'_k|^{\mathrm{SF}},$$

*with equality if and only if $e_k = 1$.*

*Proof.* Codimension-1 removal: the covering element with modulus $p_k$ (i.e. $a_k = 1$, all other coordinates free) selects a coset of $p_k^{e_k-1}$ values at coordinate $k$. Removing this coset gives $|S'_k| = p_k^{e_k} - p_k^{e_k-1} = p_k^{e_k-1}(p_k-1)$. For $e_k \geq 1$, $p_k^{e_k-1} \geq 1$, so $|S'_k| \geq p_k - 1$. □

### 3.3. **LP lifting.**

**Proposition 3.2** (Initial measure monotonicity)**.** *The LP-optimal initial measure satisfies $c'_a(x) \leq c_a(x)$ for all $x \geq 0$, where $c'_a$ and $c_a$ denote the generating functions in the non-square-free and square-free settings respectively.*

*Proof.* Let $P_a^*$ be the SF-optimal measure on $Q_a = [p_1] \times \cdots \times [p_a]$. Define the natural projection $\pi : Q'_a \to Q_a$ by $\pi(x')_i = x'_i \bmod p_i$, and lift $P_a^*$ to $Q'_a$ by $P'_a(x') = P_a^*(\pi(x'))/|\pi^{-1}(\pi(x'))|$.

A coset $C$ in $Q'_a$ with coordinate-$i$ valuation $a_i$ satisfies $P'_a(C) = P_a^*(H) \cdot \prod_{i\in I} p_i^{-(a_i-1)}$ for a corresponding hyperplane $H$ in $Q_a$. Since $a_i \geq 1$, we have $P'_a(C) \leq P_a^*(H)$, giving $c'_a(I) \leq c_a(I)$ for every index set $I$.

As $P'_a$ is a feasible solution to the non-square-free LP, the LP-optimal value is at most the lifted value, which is at most the SF-optimal value. Therefore $c'_a(I) \leq c_a(I)$ for all $I$, and hence $c'_a(x) = \sum_I c'_a(I)\, x^{|I|} \leq \sum_I c_a(I)\, x^{|I|} = c_a(x)$ for $x \geq 0$. □

### 3.4. **Combined monotonicity.**

**Theorem 3.3** (Sieve value monotonicity)**.** *For all $k \geq a$ and $x \geq 0$,*

$$c'_k(x) \;\leq\; c_k(x). \tag{5}$$

*Consequently, $M_{1,k}^{\mathrm{nonSF}} \leq M_{1,k}^{\mathrm{SF}}$ and $M_{2,k}^{\mathrm{nonSF}} \leq M_{2,k}^{\mathrm{SF}}$ for every $k$.*

*Proof.* By induction on $k$. The base case $k = a$ is Proposition 3.2. For $k > a$, the recurrence (1) gives

$$c'_k(x) \;\leq\; c'_{k-1}(x)\left(1 + \frac{x}{(1-\delta_k)\,|S'_k|^{\mathrm{nonSF}}}\right).$$

By the inductive hypothesis $c'_{k-1}(x) \leq c_{k-1}(x)$, and by Proposition 3.1 $|S'_k|^{\mathrm{nonSF}} \geq |S'_k|^{\mathrm{SF}}$, each factor on the right is at most the corresponding SF factor. Therefore $c'_k(x) \leq c_k(x)$.

For the moments: by (2)–(3), $M_{1,k}$ has $c_{k-1}(1)$ in the numerator and $|S'_k|$ in the denominator. Both move favourably: the numerator decreases (since $c'_{k-1}(1) \leq c_{k-1}(1)$) and the denominator increases. For $M_{2,k}$, the numerator

$c_{k-1}(3)-2c_{k-1}(1)+1$ also decreases, since it equals $1+\sum_{|I|\geq 1} c_{k-1}(I)(3^{|I|}-2)$ with each $c'_{k-1}(I) \leq c_{k-1}(I)$ and $3^{|I|}-2>0$. □

## 4. Proof of the main theorem

*Proof of Theorem 1.2.* Suppose for contradiction that a covering system with distinct odd moduli $n_1, \dots, n_k > 1$ exists.

**Step 1** (Setup). Construct the direct extension box $Q'$ and apply the BBMST sieve with the same distortion parameters $\delta_k$ used in the SF proof.

**Step 2** (Monotonicity). By Theorem 3.3, $M^{\mathrm{nonSF}}_{i,k} \leq M^{\mathrm{SF}}_{i,k}$ for $i = 1, 2$ and every $k$. Therefore, the sieve sum (4) satisfies

$$\sum_{k>a} \min\Big\{M^{\mathrm{nonSF}}_{1,k},\ \frac{M^{\mathrm{nonSF}}_{2,k}}{4\,\delta_k(1-\delta_k)}\Big\} \ \leq\ \sum_{k>a} \min\Big\{M^{\mathrm{SF}}_{1,k},\ \frac{M^{\mathrm{SF}}_{2,k}}{4\,\delta_k(1-\delta_k)}\Big\} \ <\ 1,$$

where the last inequality is the content of Theorem 1.1 (BBMST [1]).

**Step 3** (Conclusion). By Theorem 2.1, a sieve sum less than 1 implies that the arithmetic progressions do not cover $\mathbb{Z}$, contradicting our assumption. □

*Remark* 4.1 (On parallel cosets). When multiple moduli share the same prime set $F$, they produce "parallel cosets" at coordinate $k$ — multiple cosets contributing to $\alpha_k$. One might worry that these additional terms inflate $M_{2,k}$. However, the BBMST moment bounds (2)–(3) are *worst-case* bounds depending only on $c_{k-1}$ and $|S'_k|$, not on the number of covering elements. The monotonicity of both quantities (Theorem 3.3 and Proposition 3.1) therefore handles parallel cosets automatically, with no additional argument needed.

## 5. Lean 4 formalization

The argument has been formalized in Lean 4 with Mathlib (786 lines, compiled against Mathlib v4.28.0). The formalization contains exactly two axioms, both citing published BBMST results:

(1) `sieve_criterion` (BBMST [1, Theorems 3.1 and 3.2]): if the sieve sum (4), computed using the covering system's own effective block sizes, is less than 1, then the covering system cannot cover $\mathbb{Z}$.

(2) `sf_sieve_lt_one` (BBMST [1, Theorem 1.1]): for any set of odd primes there exist distortion parameters $\delta_k$ and LP initial values $(c_a(1), c_a(3))$ such that the square-free sieve sum is less than 1. The axiom also asserts the structural LP properties $c_a(1) > 0$, $c_a(1) \leq c_a(3)$, $c_a(3) - 2\,c_a(1) + 1 \geq 0$, and $0 \leq \delta_k < 1$.

The entire coset monotonicity argument (our contribution) is proved in Lean without `sorry`. The key results verified are:

- `sieveProd_antitone`: the sieve product $\prod(1 + x/((1-\delta_k)\,s_k))$ is antitone in the block sizes $s_k$.

- `sieveProd_diff_mono` (*core lemma*): the difference $P_1(x) - P_2(x)$ between sieve products for two block-size configurations ($s_1 \le s_2$ componentwise) is monotone increasing in $x \ge 0$. The proof is by induction on the list of block sizes, decomposing at each step into a product of nonnegative increasing factors.
- `M2_num_antitone`: the $M_2$ numerator $c_a(3)\,P(3) - 2\,c_a(1)\,P(1) + 1$ is antitone in block sizes. This uses `sieveProd_diff_mono` together with the algebraic identity $\Delta u_3 = 3\,\Delta u_1$ (the update-factor difference at $x = 3$ is three times the difference at $x = 1$) and the bound $3\,c_a(3) > 2\,c_a(1)$.
- `sieveSum_antitone`: the full sieve sum with non-square-free block sizes is at most the sieve sum with square-free block sizes.
- `blockSize_effective_le`, `weight_le_sf`, `weight_strict`: block size and coset weight comparisons.
- `erdos_selfridge`: the main theorem, connecting the monotonicity chain to the two axioms.

The command `#print axioms erdos_selfridge` reports only the two declared axioms together with the standard Lean axioms (`propext`, `Classical.choice`, `Quot.sound`).

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## References

[1] P. Balister, B. Bollobás, R. Morris, J. Sahasrabudhe, and M. Tiba, *The Erdős–Selfridge problem with square-free moduli*, Algebra & Number Theory **15** (2021), no. 3, 609–626. `arXiv:1901.11465`.

[2] P. Erdős, *On integers of the form $2^k + p$ and some related problems*, Summa Brasil. Math. **2** (1950), 113–123.

[3] P. Erdős, *Some recent advances and current problems in number theory*, in: Lectures on Modern Mathematics III (T. L. Saaty, ed.), Wiley, New York, 1965, pp. 196–244.

[4] B. Hough, *Solution of the minimum modulus problem for covering systems*, Ann. of Math. (2) **181** (2015), no. 1, 361–382.

[5] R. D. Hough and P. P. Nielsen, *Covering systems with restricted divisibility*, Duke Math. J. **168** (2019), no. 17, 3261–3295.

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